Machine Learning

Deep learning:
Deep neural networks

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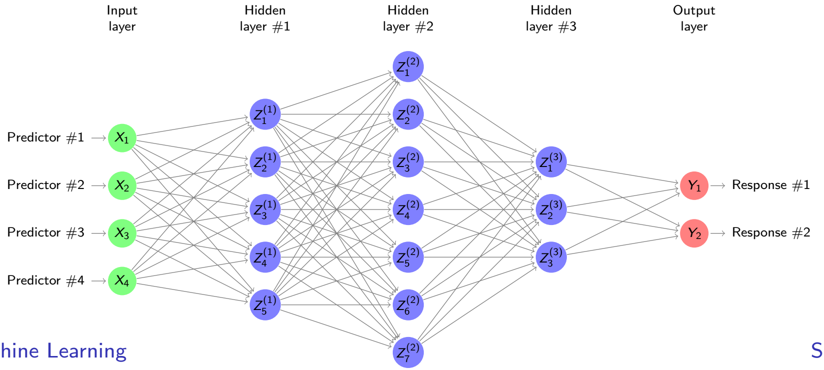
Deep neural networks

- A deep neural network is a neural network with multiple hidden layers between the input and the output layers.
- The predictive model is then defined sequentially

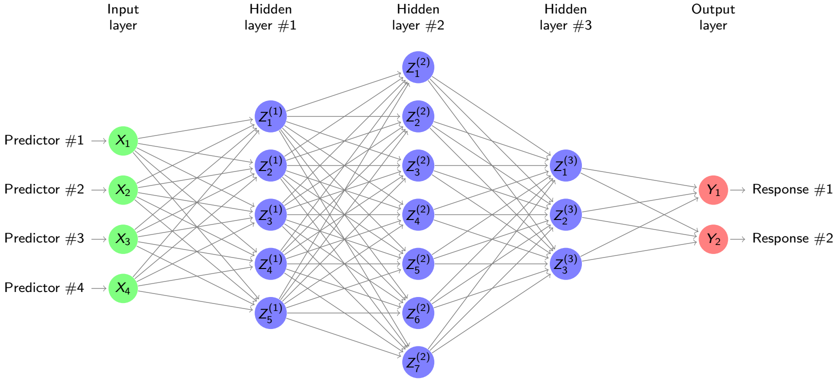
$$
\begin{aligned}
Z^{(1)} &= g_1(b^{(1)} + \mathbf{W}^{(1)} X), \\
Z^{(2)} &= g_2(b^{(2)} + \mathbf{W}^{(2)} Z^{(1)}), \ldots, Z^{(M)} = g_M(b^{(M)} + \mathbf{W}^{(M)} Z^{(M-1)}) \\
\widehat{f}_i(X) &= h_i(b^{(M+1)} + \mathbf{W}^{(M+1)} Z^{(M)}), \quad i = 1, \ldots, q,
\end{aligned}
$$

with weight matrices $\mathbf{W}^{(1)}, \ldots, \mathbf{W}^{(M+1)}$, bias vectors $b^{(1)}, \ldots, b^{(M+1)}$, and functions $g_1, \ldots, g_M, h_1, \ldots, h_q$.



The architecture of deep neural networks

- The `depth` of the neural network is the total number of layers.
- For each layer we call the number of neurons the `width` of that layer.



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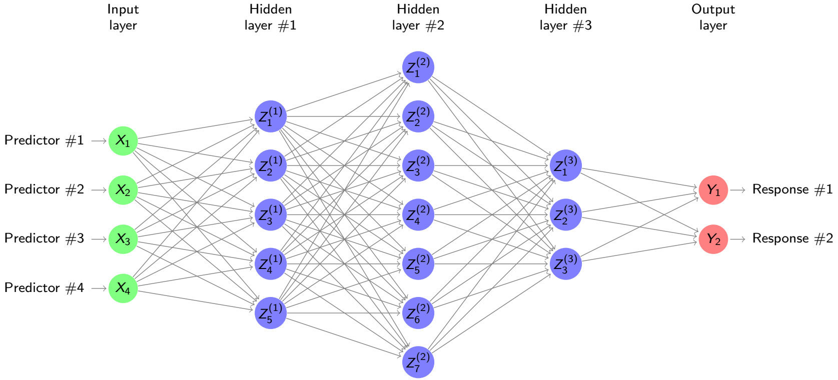

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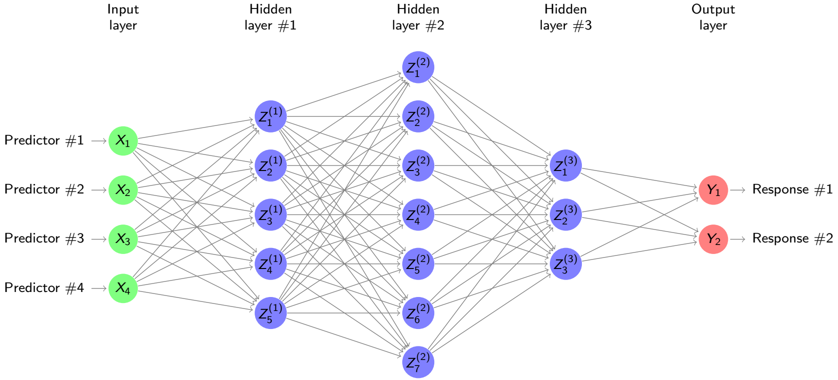
- A good architecture is based on experience and cross-validation.



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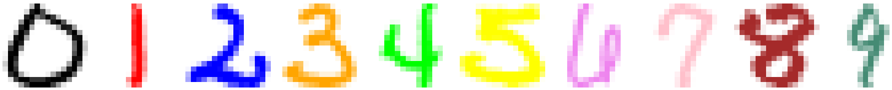
- Deep neural networks have `many (!) parameters`, meaning that fitting them is computationally `very costly` and `overfitting` must actively be prevented.

A simple example: digit classification



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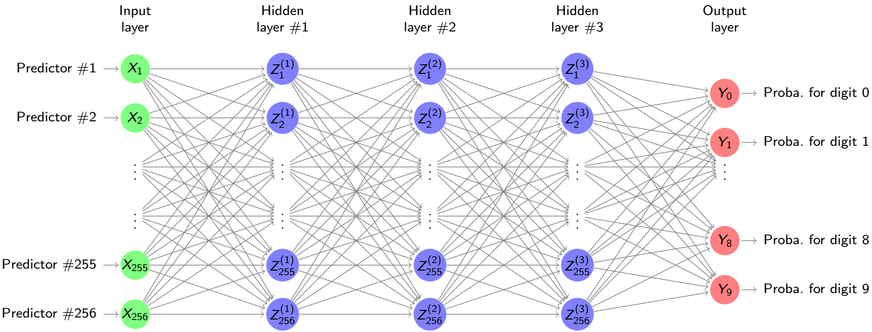
- Consider a `fully connected` network with $M = 3$ hidden layers, each having 256 neurons.



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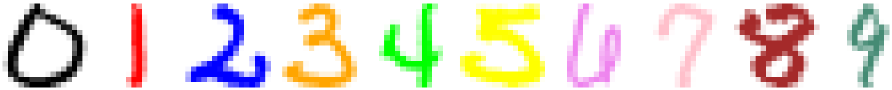

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- This "simple network" already has $3 \times 257 \times 256 + 257 \times 10 \approx 200k$ parameters!